



RANGE

Why Generalists Triumph
in a Specialized World

DAVID EPSTEIN



The Big “So What”

In this book, Epstein challenges conventional wisdom that success is best achieved through hyperspecialization, and children should be nurtured in a specific field as early as possible to give them a head-start. He uses stories and research studies to show why/how “range” can be equally (if not more) critical for long-term success in a complex world.



Introduction

➡ The book opens with the stories of 2 sports superstars—Tiger Woods and Roger Federer—who represent specialization vs range.

- Tiger Woods’ father saw him as a golf prodigy. He began training Tiger at the young age of 2, and devoted his life to nurturing his son into a golf superstar. Tiger is the perfect example of specialization—selecting a path early and accumulating hours of targeted training to become outstanding in an area.
- Roger Federer became a tennis legend even though he started much later. When he was young, he tried all sorts of sports, e.g. skiing, swimming, wrestling, skateboarding, basketball, badminton, soccer, etc. He was talented in tennis, but only chose to focus on the sport when he was 12 years old.
- Epstein argues that Roger’s circuitous path to success is common but less publicized. There’re many other sports icons who tried different sports before settling on one (e.g. Tom Brady, Nick Foles) and some even mastered

KEY QUOTES

“In the wicked world, with ill-defined challenges and few rigid rules, range can be a life hack.”

“Tiger has come to symbolize the idea that the quantity of deliberate practice determines success—and its corollary, that the practice must start as early as possible.”

multiple sports (e.g. Ester Ledecká who won gold for skiing *and* snowboarding at the same Olympics).

 **Specialization vs range.** We're often told that high achievers stick to one career path and persist till they succeed. We frown upon those who keep changing jobs.

- Hyperspecialization is about focusing on 1 area and using deliberate practice to hone your skills. It suggests that there's a linear path from A to B, and the earlier you train in a specific domain, the higher your chances of success.
- Epstein argues that specialization is overrated. Instead, he makes the case for "range": to accumulate diverse skills/experiences that culminate in a late specialization that fits with your unique strengths and preferences. People who pursue this path tend to go through a relatively unstructured "sampling period" where they try different things, develop various skills and learn about themselves.
- He suggests that both specialization and range are essential for progress and fulfillment. In fact, range can be even more valuable when we encounter complex, unprecedented problems.
- The book is organized into 12 key chapters. Epstein doesn't directly state the takeaways, but uses stories and examples to broadly suggest why/how wide-sampling and late-specialization work. In this summary, we've distilled the main ideas and examples into 12 sections/themes below.

KEY QUOTES

"Mental meandering and personal experimentation are sources of power, and head starts are overrated."

*"We know that early sampling is key, as is diversity."
—Ross Tucker*



The Case for Developing Range



Head-starts are overrated

 There're many stories of iconic figures who had a head-start since young, e.g. Mozart with music, Mark Zuckerberg with computers. Laszlo Polgar also took the world by storm when he "experimented" with his 3 daughters and successfully trained/home-schooled them to become champion chess players.

- Stories like Polgar's and Tiger Woods' advocate that you start as early as possible and use deliberate practice to develop your skills. However, chess and golf are not representative of all critical skills/domains in life.
- Psychologists Gary Klein and Daniel Kahneman independently studied the relationship between experience and expertise. They found that experience didn't always lead to expertise, especially where human behavior and patterns weren't clearly repetitive.

 The key lies in the *nature* of the environments/challenges. Psychologist Robin Hogarth calls these kind vs wicked learning environments.

- Kind learning environments refer to situations where (i) patterns are repeated regularly, and (ii) feedback is both immediate and accurate. Certain games, music and sports (including golf and chess) occur in kind learning environments: they have clearly-defined rules/boundaries, and observable cause-effect outcomes that allow learners to improve through repetition.
- Wicked learning environments refer to situations where (i) the rules of game are unclear/incomplete, (ii) there may or may not be repetitive/obvious patterns, and (iii) feedback tends to be delayed and/or inaccurate. In wicked environments (e.g. financial markets), past patterns don't predict the future and experience may teach you the wrong lessons.
- Experts are strong in their domain because they've learned to recognize patterns in split seconds. Those advantages disappear if they're placed in a new environment, or if there's a change in status quo. For example, an elite footballer may appear to have superhuman reflexes because he recognizes patterns in ball/body movements to help him to anticipate what'll happen. This reflex disappears when he plays a new/unfamiliar sport.

 In wicked environments, narrow specialization can even become a disadvantage.

- Experts may get so used to certain approaches that they stick rigidly to them even when the "proven" methods no longer work.

KEY QUOTES

"Whether or not experience inevitably led to expertise... depended entirely on the domain in question."

"Kind learning environment experts choose a strategy and then evaluate; experts in less repetitive environments evaluate and then choose."

"As complexity increases... we also need more people... with range."

- It's also easier to replicate a narrow specialization than broad know-how. So far, artificial intelligence (AI) has been able to trump humans in narrow, tactical areas with clearly-defined parameters. For example, machines have already beaten human experts in chess—which is orderly, rule-bound and mainly tactical. However, in undefined, open-ended scenarios with imperfect data and a lack of clear rules (e.g. a game of warcraft), humans still have an edge due to our ability to think strategically and integrate information creatively.

 In short, for every person who succeeds with a head-start, you can find a late-bloomer who succeeds without it. Balance the need for specialist knowledge with the risk of rigid thinking:

- Vary the challenges within your domain. Don't assume that old patterns will remain the same.
- Broaden your perspective by keeping 1 foot outside your domain at all times. Draw on outside experiences and analogies, and deliberately apply the knowledge from one domain to another.



Complexity in the modern world

 Professor James Flynn found that our average IQ scores have increased by 3 points every decade in the 20th century, with improvements mainly in the areas of abstract thinking, problem solving and reasoning. This was supported by other studies:

- Psychologist Alexander Luria found that people in modern societies think differently from those in pre-modern villages. For example, the villagers could only grasp tangible concepts that related directly to their daily activities. When they were presented with a hammer, a saw, a hatchet, and a log, they couldn't see how the first 3 items could be "tools" since they were useless without the log. They also couldn't see how 4 dotted lines could form a square.
- People in post-industrialization societies are much more flexible in our thinking and can use layers of abstract

KEY QUOTES

"Our greatest strength is the exact opposite of narrow specialization. It is the ability to integrate broadly."

"The challenge we all face is how to maintain the benefits of breadth, diverse experience, interdisciplinary thinking, and delayed concentration in a world that increasingly incentivizes, even demands hyperspecialization."

"[Premodern villagers] were perfectly capable of learning from experience, but failed at learning without experience."

"Great rewards will accrue to those who can take conceptual knowledge from one problem or domain and apply it in an entirely new one."

concepts and categories to develop complex models. The word "percent" was barely found in books in 1900, but appeared once in every 5000 words by the year 2000. Computer software and digital devices are all built on abstract concepts and symbols.

- Increasingly, standard and repetitive problems will be automated and solved by machines. On the other hand, complex problems will require the ability to think fluently across domains.



To solve complex problems in the modern world, you need (i) a range of tools and experience, as well as (ii) the ability to think critically and integrate ideas across domains. Learn how to think, not just *what* to think.



Wide-sampling for learning and creativity



High-achievers don't succeed overnight. They, too, have to go through phases of learning and discovery.

- The figlie del coro was a group of female musicians in Venice known for their musical versatility. The best of them (e.g. Anna Maria della Pietà) became celebrities across Europe. Figlie were actually orphans taken in by the orphanage Ospedale della Pietà. They spent the bulk of their time on chores and regular work. However, during their formal music lessons (3x a week) and music study (1 hour a day), they learned to sing and play *every* instrument in the institution. This ability to play multiple instruments was core to their musical skills and versatility. When famous musician Pelegrina della Pietà lost her teeth in her sixties, she had to give up the oboe but could continue to perform on the violin.
- Many paths to music excellence actually start with a sampling period where the student has lightly-structured lessons and is exposed to a range of activities/instruments. Focused, highly-structured, intense practice comes later. In fact, the most accomplished students typically play at least 3 instruments. For example, famous cellist Yo-Yo Ma moved from the violin to the piano before settling on the cello.

KEY QUOTES

"Everyone needs habits of mind that allow them to dance across disciplines."

"The sampling period is not incidental to the development of great performers...it is integral."

 When skills are learned/applied across a range of contexts, it develops abstract mental models that learners can apply in new situations. *That is the root of creativity.*

- Jazz music is much less structured than classical music. A classical piece may be perfected through practice (by repeatedly identifying and correcting errors), whereas jazz music requires improvisation and creativity. That's why it's generally easier for a jazz musician to learn classical music than vice versa. The improvisation process is similar to how babies learn to talk—they pick up the sounds *before* learning about grammatical rules.
- Django Reinhardt learned music from observation and trial-and-error. At 18, he was injured in an accident and had to try new ways to play without using his left pinkie and ring finger. This sparked a burst of creativity. Reinhardt never learned to read music, yet he could compose an entire symphony by playing each instrument part on his guitar for another musician to transcribe.



Fast vs slow learning

 Parents and teachers are often overly-focused on instant results. In reality, the most effective learning strategies are slow and effortful.

- In pedagogy, there are 2 key types of questions:
 - (i) Procedural questions that help students to practice something they've learned, e.g. applying a formula to different math problems.
 - (ii) Connection questions that help students to grasp a broader concept, e.g. why a formula works in different situations.
- In both cases, the students must figure out the answers on their own for the insights to "stick". When teachers offer hints, they actually hinder the students' ability to learn deeply.

 Specifically, learning strategies that create "desirable difficulties" have been proven to enhance long-term learning:

KEY QUOTES

"The more context in which something is learned, the more the learner creates abstract models, and the less they rely on any particular example."

*"Improv masters learn like babies: dive in and imitate and improvise first, learn the formal rules later."
—Jack Cecchini*

*"What you want is to make it easy to make it hard."
—Nate Kornell*

"Frustration is not a sign you are not learning, but ease is."

- Interleaving is about mixing up the topics instead of studying them sequentially in blocks. For example, instead of repeatedly studying 1 artist's works, it's better to intermix works from several artists. This helps you to discern their differences, so you can learn to match the right strategies to different problems.
- Spacing (or distributed practice) is about splitting your practice into a few sessions instead of cramming everything into 1 session. It's harder to recall something when you return to it after some time, which is precisely why it's more effective for encoding the information into your long-term memory.
- The generation effect is about testing yourself and generating answers on your own (even if they're wrong). The most effective learning happens when an answer you're confident about turns out to be wrong.

✓ In short, **learning that's challenging, slow and frustrating in the short-term tends to yield better results in the long-term.** On the other hand, short-cuts that boost immediate performance (e.g. hints, cramming) tend to undermine long-term results.

Handling never-seen-before situations

 Deep analogical thinking is about identifying conceptual similarities across seemingly-different situations. This helps us to think through and understand a problem that we have no experience with.

- Johannes Kepler's Laws of Planetary Motion gave birth to modern astrophysics. At that point, gravity hadn't been discovered yet, and it was still widely-believed that the universe's planets moved at a constant speed around Earth. Each time he got stuck, Kepler used analogies to consider a new perspective. For example, he likened planets to "moving souls", imagined how "light-like forces" could cause planetary motion, and how planets could attract/repel one another like "magnets".

KEY QUOTES

"For knowledge to be flexible, it should be learned under varied conditions."

"Short-term rehearsal gave purely short-term benefits... Repetition, it turned out, was less important than struggle."

"Learning is most efficient in the long run when it is really inefficient in the short run."

"Analogical thinking takes the new and makes it familiar, or takes the familiar and puts it in a new light, and allows humans to reason through problems they have never seen in unfamiliar contexts. It also allows us to understand that which we cannot see at all."

"In a wicked world, relying upon experience from a single domain is not only limiting, it can be disastrous."

- The more analogies you consider, the more you're likely to arrive at a creative solution. Try this famous cognitive exercise: Duckner's radiation problem.
- Imagine you're a doctor. Your patient has a tumor that can't be operated upon. You can use a certain ray to destroy the tumor. However, it must be applied at a high level of intensity which will also destroy the healthy tissues that it passes through. What can you do?
- Most people are stumped by this challenge initially, but can solve it after being given 2 analogies. First, a large army was split into groups so they could take many small roads to converge at the fortress they were attacking. Second, many villagers put out a big fire by simultaneously pouring a bucket of water each into the fire. With these analogies, the solution to the tumor may suddenly seem obvious: use multiple rays of lower intensity to target the tumor from different angles. This will produce a high-enough combined intensity to destroy the tumor without killing the surrounding tissues.

 Inside vs outside views. Studies have found that humans make poor judgements when we focus narrowly on details in front of us. The more internal details you have about a scenario (e.g. the physical qualities of a race-horse, or the background/details of a politician running for election), the more likely you'll predict that the scenario will occur. By contrast, when you explore outside views, examine structural similarities between your scenario and other remote scenarios, you make better judgements.

 In short, you can use analogical thinking to address new, unfamiliar challenges. Good analogical thinking requires that you (i) refer to multiple examples from different and distant domains, and (ii) look beyond superficial/obvious features to identify the deep structural similarities.



Know when to quit

 Late-specialization isn't bad. In fact, it may be vital for match quality—the fit between your work and your abilities/dispositions.

KEY QUOTES

"A problem well put is half-solved."
—John Dewey

"Successful problem solvers are more able to determine the deep structure of a problem before they proceed to match a strategy to it."

- Van Gogh spent the bulk of his life exploring and failing in many areas—as a student, art dealer, bookseller, teacher, preacher and aspiring pastor. At the age of 33, he enrolled in art school, but dropped out and developed his own artistic style. When he passed away at 37, his works were only just starting to be recognized. He became truly famous only after his death. Likewise, other late bloomers (e.g. Gauguin and J.K. Rowling) thrived in spite of (or possibly *because of*) their late start.
- After studying of thousands of college graduates, economist Ofer Malamud found that match quality improved when students had more sampling opportunities. English and Welsh students specialized early since they had to apply for a specialized program to enter college. On the other hand, Scottish students could spend their first 2 years studying different fields and continue sampling thereafter. The results showed that: (i) English and Welsh graduates were consistently more likely to leave their career fields compared to their Scottish peers, and (ii) Scottish graduates started out with fewer specialized skills and a lower salary, but quickly caught up.

 Having said that, isn't persistence important for career success? Psychologist Angela Duckworth found that grit (a combination of passion and perseverance) was a better predictor of whether cadets would complete the "Beast Barracks"—the West Point's training program—compared to the military's Whole Candidate Score.

- Epstein pointed out that the Army's goal was to nurture career senior officers, not Beast survivors. Unfortunately, about half of West Point graduates were leaving active military service after 5 years (once they were allowed to). West Point professors believed there could be an issue with match quality: high-potential candidates may perform well in the military, but they also had great career options outside the military, and ultimately chose to pursue those options.
- In short, grit isn't always good. It may predict whether you complete a course or win a contest, but it can become an obstacle to match quality. **Instead of persisting on every task/path, sometimes it's good to quit if you're clearly on the wrong track.** Just don't quit as an easy way out.

KEY QUOTES

"Exploration is not just a whimsical luxury of education; it is a central benefit."

"If we treated careers more like dating, nobody would settle down so quickly."

*"We fail when we stick with tasks we don't have the guts to quit."
—Seth Godin*

"The important trick...is staying attuned to whether switching is simply a failure of perseverance, or astute recognition that better matches are available."



Experiment to discover your possible selves

 We're often told that we should have a clear vision for the future, then work backwards to develop a plan. For career successes, a short-term, trial-and-error approach may work better.

- Frances Hesselbein (former CEO of the US' Girl Scouts, then CEO of Peter Drucker's leadership institute) is one of the most respected CEOs of her time. She didn't have formal business training, but is always open to new ideas/experiences, and constantly seeks to learn something from each new experience.
- Hesselbein's journey is not uncommon. The Dark Horse Project found that many successful people had circuitous career paths instead of direct, linear ones. They use short-term planning to leverage available opportunities that match who they were *at that time*. They only developed long-term goals after periods of self-discovery.

 So, instead of rigidly following a fixed plan, it's better to continually uncover new possibilities by experimenting in different domains. This makes sense because:

- Our traits/characteristics change depending on the context. For example, a child can be passive at home but aggressive in school.
- We also change with age and experience, even though we can't predict how we'll change. Psychologist Dan Gilbert uses the term "end of history illusion" to describe how humans are able to retrospectively make sense of how we've changed in the past, but are unable to conceive our future changes.
- Professor Herminia Ibarra recommends using real-life experiences to test and learn who you are, rather than relying on personality tests and introspection to choose your ideal career.

KEY QUOTES

*"First act and then think. We discover the possibilities by doing, by trying new activities, building new networks, finding new role models."
—Herminia Ibarra*

"We learn who we are in practice, not in theory."

"The precise person you are now is fleeting, just like all the other people you've been."

*"Test-and-learn, not plan-and-implement."
—Herminia Ibarra*



Tap on the outsider advantage



Specialists may not always have the best solutions to complex, technical problems. Outsiders can have an advantage, regardless of whether they have any domain knowledge.

- InnoCentive is an open-innovation and crowdsourcing platform that posts challenges for the public to solve (usually for a prize). Using this platform, the Alaska-based Oil Spill Recovery Institute found a simple, practical solution for the oil spills that had contaminated Alaska's coasts for almost 20 years. The solution came from John Davis, a chemist who recalled how he used a vibrating rod to keep concrete from solidifying, and proposed a similar solution to whip up the oil spills so they could be removed.
- Jill Viles suffered from a genetic disease that led to muscular dystrophy, but doctors couldn't explain why the gene affected her more severely than her family members. Viles researched widely and managed to establish that she had Emery-Dreifuss (with a mutation of the lamin gene). Since Emery-Dreifuss was linked to heart problems, she quickly sought help for her father's irregular heartbeats and ended up saving his life. Several years later, she noticed that sprinter Priscilla Lopes-Schliep had an unusual lack of fat on her limbs, like herself (except Priscilla had much more muscle). She managed to connect with Priscilla and prove her hypothesis that they both had partial lipodystrophy (another rare disease). In fact, Priscilla could've suffered a pancreatitis attack if her condition had not been highlighted. Through her own observations and research, Viles uncovered answers and saved lives in a way specialists couldn't.



We can facilitate discoveries by creating opportunities for specialist information to be made available to outsiders. The latter is in a better position to unlock creative solutions by (i) reframing the problem with a fresh perspective, and (ii) synthesizing the ideas across vastly-different fields.

KEY QUOTES

"Knowledge is a double-edged sword. It allows you to do somethings, but it also makes you blind to other things that you could do."
—Pedro Domingos

"Big innovation most often happens when an outsider who may be far away from the surface of the problem reframes the problem in a way that unlocks the solution."
—Karim Lakhani



Combining breadth and depth

Epstein uses 3 stories to share ideas about creativity.



Nintendo's engineering strategy is largely shaped by Gunpei Yokoi. Yokoi knew he had limited engineering capabilities and couldn't fight head-on with competitors' cutting-edge technology. So, he used "lateral thinking with withered technology", i.e. to re-imagine and combine old technology (that's cheap, widely available and understood) in new and creative ways. For example, screens from common LCD calculators were used to create a portable handheld game which led to the popular Game Boy. In an industry that focuses on specialized know-how, Nintendo succeeds by using existing technology in fresh ways. Yokoi believed that lateral and vertical thinkers are both valuable, and are the strongest when they work together.



One of 3M's top inventors, Andy Ouderkirk, set out to identify the ideal inventor profile. He found 3 types of inventors at 3M. *Specialists* focus on a narrow area (e.g. a specific type of plastics) and tend to have patents only in that area. *Generalists* integrate ideas from different domains and tend to have patents in diverse areas. *Polymaths* are a hybrid between generalists and specialists—they have a broad range and at least 1 area of depth.

- Specialists thrive in problems that're well-defined and well-understood, but generalists and polymaths do better where there's more ambiguity and uncertainty. Polymaths are the most likely to win the esteemed Carlton Award.
- To facilitate the cross-fertilization of ideas, create opportunities for specialists to develop range. Ouderkirk himself was a chemist, but acquired a breadth of experience when he worked with >100 teams in his first 8 years at 3M.



In a study of what made comic books successful, the researchers found that breadth of experience matters, especially when it resides in 1 individual.

- The creators' breadth of experience mattered more than the length of experience, as the former affected the level and quality of innovation.

KEY QUOTES

"Specialization is obvious: keep going straight. Breadth is trickier to grow."

"Seeding the soil for generalists and polymaths who integrate knowledge takes more than money. It takes opportunity."

"A diverse group of specialists cannot fully replace the contributions of broad individuals."

- An individual creator with experience in ≥ 4 genres was more innovative than a team whose members jointly have experience in ≥ 4 genres. This could be because it's easier for an individual to integrate diverse experiences in his/her head.



Don't be fooled by expertise



In 1984, psychologist and political scientist Philip Tetlock began to collect thousands of predictions by 284 experts. He found that most experts were terrible forecasters, regardless of their specialty, experience, or academic degrees.

- Expert forecasts were consistently wrong—both for short-term and long-term forecasts, and across all domains. Events that were declared impossible (or nearly impossible) still happened 15% of the time. Things that were supposed to happen with certainty failed to happen $>25\%$ of the time. In fact, the more famous the expert, the more likely they were wrong.
- By contrast, integrators who researched and combined the insights from various experts generated better forecasts on most fronts, especially for long-term predictions.
- Tetlock called them “foxes” vs “hedgehogs”. Generally, foxes who knew many things tended to outperform hedgehogs, because the former could combine different viewpoints into their analyses, allowing them to calibrate their predictions more accurately.



Tetlock also identified several characteristics of great forecasters:

- The best forecasters were both (i) foxy individuals who were curious about everything and had a wide range, and (ii) effective collaborators who actively shared information and discussed predictions. They saw their ideas as hypotheses to be tested—instead of convincing others that they were right, they invited others to prove them wrong, thus accelerating their team's learning.

KEY QUOTES

“Good judges are good belief updaters. If they make a bet and lose, they embrace the logic of a loss just as they would the reinforcement of a win.”

—Philip Tetlock

“The foxiest forecasters were impressive alone, but together they exemplified the most lofty ideal of teams: they became more than the sum of their parts. A lot more.”

- To improve your forecasting abilities, don't just focus on internal details of the event being investigated. Instead, (i) examine distinctly different events with deep structural similarities, and (ii) study failed predictions to learn what went wrong.



Know when to drop familiar tools



The Carter Racing case study is widely used in MBA programs. Teams are told the importance of an upcoming race, and given data (including the ambient temperature) of 7 engine failures that occurred out of 24 races. Although the last 2 races went smoothly after some fixes, the team didn't know if the problem was resolved. They now have a short time to decide whether to proceed with the race.

- In the real-world, we often have to make important decisions under time pressure, with incomplete information. In fact, this case study was designed based on NASA's 1986 Challenger shuttle explosion. A key question was whether ambient temperature affected engine failure. The data for the 7 failures would've suggested that temperature didn't matter, and students invariably made their decision based on this data. If they had asked for additional data for races *without* engine failures, they'd have seen a clear pattern: engine failures had occurred in every race below 65F, meaning a 99.4% chance of engine failure at the present temperature. **Rather than focusing only on the data in front of you, you must consider what vital data may be missing.**



When we're under pressure, we tend to regress to habitual responses (e.g. methods, tools, actions) that we're the most comfortable with. This could lead to disastrous results.

- For example, firefighters are so used to their equipment that they won't drop it even when they're fleeing from an out-of-control fire. Bugged down by the heavy gear, many firefighters perish.
- Likewise, every organization has certain cultural responses that people default to. In NASA's Challenger tragedy, one of the engineers had argued strongly against the launch.

KEY QUOTES

"[The Carter Racing case study is]...a stark lesson in the danger of reaching conclusions from incomplete data, and the folly of relying only on what is in front of you."

*"When you don't have any data, you have to use reason."
—Richard Feynman*

Unfortunately, his views were dismissed as he didn't have the quantitative data to back up his claims. The NASA team was so used to thinking quantitatively that they ignored the qualitative evidence.

- To build a more effective problem-solving culture, you must become aware of your default/standard practices, then develop forces that'd push you in the opposite direction. This helps to expand your range and make it easier to drop irrelevant habits/tools when you face unfamiliar challenges.



Be a deliberate amateur

 Best-practices and guidelines are useful, but don't allow them to restrict you. Epstein argues that the most creative thinkers tend to have a playful streak that allows them to explore freely and transcend conventional boundaries.

- Nobel Prize winner Andre Geim uses "Friday night experiments" to try out random, crazy ideas. Most don't amount to anything, but a few sparked ideas that led to serious innovations. For example, his attempt to use Scotch tape to rip thin layers of graphite led to the development of graphene: a material 100,000x thinner than human hair and 200x stronger than steel.
- Don't be too serious or perfectionistic. Occasionally allow yourself to slip into "limited sloppiness", e.g. infuse a different style into your work even if it's "improper". Pretend you're a beginner, or intentionally veer off your original path to see what may surface.
- After focusing deeply on a subject/area, take time out to study other topics, so you don't become limited to a narrow sphere.
- In reality, we can't predict what type of information will lead to future breakthroughs. Instead of insisting on immediate, concrete applications for every discovery, approach your experiments with a spirit of discovery and curiosity.

KEY QUOTES

"There are no tools that cannot be dropped, reimagined, or repurposed in order to navigate an unfamiliar challenge...it is, of course, easier said than done. Especially when the tool is the very core of an organization's culture."

"An enthusiastic, even childish, playful streak is a recurring theme in research on creative thinkers."

*"Truly original discoveries in science are often triggered by unpredictable and unforeseen small findings."
—Yoshinori Ohsumi*

*"Be careful not to be too careful or you will unconsciously limit your exploration."
—Max Delbruck*